Appendix A. The procedure of the label-setting algorithm

The procedure of the label-setting algorithm is shown in Algorithm A.

Algorithm A. Label-setting algorithm

1. Let $\mathcal{H}_i$ be the set of labels with the last node i. $\mathbb{E}$ is the set of current nodes for the expanded labels. $\mathcal{P}$ is the set of columns with negative reduced costs.
2. $\mathcal{H}_0 \leftarrow \{L_0\}, \mathcal{P} \leftarrow \emptyset, \mathbb{E} \leftarrow \{0\}$.
3. **While** $\mathbb{E} \neq \emptyset$ and $|\mathcal{P}| \leq |\mathcal{I}|$ **do**
4. Select $i \in \mathbb{E}, \mathbb{E} \leftarrow \mathbb{E} \backslash \{i\}$.
5. **For** all $L_i \in \mathcal{H}_i$ **do**
6. Obtain $\tilde{\mathcal{O}}$ (set of nodes to be expanded) via Algorithm B (see Appendix B).
7. **For** each $j \in \tilde{\mathcal{O}}$ **do**
8. $L_j \leftarrow$Extend (L_i, j).
9. **If** $j \in \mathcal{B} \cup \{n+1\}$ **then**
10. **If** $\overline{c}(L_j) < 0$ **then**
11. $\mathcal{P} \leftarrow \mathcal{P} \cup \{L_j\}$.
12. **Else**
13. **If** L_j is not dominated by either of the two enhanced dominance
14. rules or dynamic label dominance strategy II **then**
15. $\mathcal{H}_j \leftarrow \mathcal{H}_j \cup \{L_j\}, \mathbb{E} \leftarrow \mathbb{E} \cup \{j\}$.
16. Return $\mathcal{P}$.

Appendix B. The procedure of the dynamic label dominance strategy I

The procedure of the dynamic label dominance strategy I is shown in Algorithm B.

Algorithm B. Dynamic label dominance strategy I

Let L_i be the label currently at node *i*. $\mathcal{O}'(L_i)$ denote the set of nodes that can be reached subsequently by label L_i. ρ is equal to ρ_1 in the dedicated vehicle pricing problem and ρ_2 in the crowdsourced vehicle pricing problem. Let n, v, c_t be temporary variables.
$\tilde{\mathcal{O}} \leftarrow \emptyset$, $n \leftarrow 0$, $v \leftarrow \infty$.
If length $(L_i) \leq 2$ **then**
    Return $\mathcal{O}'(L_i)$.
For each $j \in \mathcal{O}'(L_i)$ **do**
    If j is the destination node **then**
        Continue.
    If $v \neq \infty$ **then**
        If $\rho d_{ij} + v \geq \rho d_{nj}$ **then**
            Continue.
    $c_t \leftarrow \rho d_{ij}, k \leftarrow true$.
    For m from length (L_i)-1 to 1 **do**
        $\overleftarrow{n} \leftarrow \mathcal{U}(L_i)[m-1], \vec{n} \leftarrow \mathcal{U}(L_i)[m]$.
        $c_t \leftarrow c_t + \rho d_{\overleftarrow{n}\vec{n}} - \pi_{c(\vec{n})}$.
        If $c_t \geq \rho d_{\overleftarrow{n}j}$ **then**
            $n \leftarrow \overleftarrow{n}, v \leftarrow c_t - \rho d_{ij}, k \leftarrow false$.
            Break.
    If $k = true$ **then**
        $\tilde{\mathcal{O}} \leftarrow \tilde{\mathcal{O}} \cup \{j\}$.
Return $\tilde{\mathcal{O}}$.

Algorithm B is used to filter nodes $j \in \mathcal{O}'(L_i)$ that cannot form a dominant label in the next expansion from the $\mathcal{O}'(L_i)$ of label L_i. First, a dominance judgment is performed based on cached information (n and v). If the dominance rule is not satisfied, traverse the nodes that L_i has expanded from back to front and constructed multiple virtual labels for re-evaluation. When any virtual label shows dominance, stop traversing, update the information of the virtual label to historical information, and do not expand node j in the next expansion of L_i.

Appendix C. Proofs

Appendix C.1. Proof of Proposition 1

Proof. Given two labels, L_i and L_j that satisfy conditions (50)−(55), to prove that L_i dominates L_j, it suffices to show that, for any subsequent feasible extension of L_j, label L_i can perform the same extension at a cost less than or equal to that of L_j. First, consider the case where both labels are extended once. Since $\mathcal{O}(L_i) \supseteq \mathcal{O}(L_j)$, any next expandable node of L_j can also be expanded by L_i. When L_i and L_j are extended to any node $\sigma \in \mathcal{O}(L_j)$, then

$$w(L_i \oplus \sigma) = w(L_i) + w(\sigma) \le w(L_j) + w(\sigma) = w(L_j \oplus \sigma), \tag{C.1}$$

$$v(L_i \oplus \sigma) = v(L_i) + v(\sigma) \le v(L_j) + v(\sigma) = v(L_j \oplus \sigma), \tag{C.2}$$

$$\mathcal{O}(L_i \oplus \sigma) = \mathcal{O}(L_i) \backslash \varsigma_\sigma \backslash \{\sigma\} \supseteq \mathcal{O}(L_j) \backslash \varsigma_\sigma \backslash \{\sigma\} = \mathcal{O}(L_j \oplus \sigma), \tag{C.3}$$

$$t(L_i \oplus \sigma) = \max\{\mathrm{e}_\sigma, t(L_i) + \mathrm{s}_{i\sigma}\} \le \max\{\mathrm{e}_\sigma, t(L_i) + \Delta_{ij}^t + \mathrm{s}_{j\sigma}\} \le \max\{\mathrm{e}_\sigma, t(L_j) + \mathrm{s}_{j\sigma}\} = t(L_j \oplus \sigma), \tag{C.4}$$

$$d(L_i \oplus \sigma) = d(L_i) + d_{i\sigma} \le d(L_i) + \Delta_{ij}^d + d_{j\sigma} \le d(L_j) + d_{j\sigma} = d(L_j \oplus \sigma), \tag{C.5}$$

$$\overline{c}(L_i \oplus \sigma) = \overline{c}(L_i) + \rho \cdot d_{i\sigma} - \pi_{c(\sigma)} \le \overline{c}(L_i) + \rho \cdot (\Delta_{ij}^d + d_{j\sigma}) - \pi_{c(\sigma)} \le \overline{c}(L_j) + \rho \cdot d_{j\sigma} - \pi_{c(\sigma)} = \overline{c}(L_j \oplus \sigma), \tag{C.6}$$

where ρ is taken as ρ_1 in the pricing problem for dedicated vehicles and as ρ_2 in the pricing problem for crowdsourced vehicles.

Thus, if $L_j \oplus \sigma$ is feasible, Eqs. (C.1)−(C.6) ensure that $L_i \oplus \sigma$ also satisfies resource constraints, and the basic dominance rules hold between $L_i \oplus \sigma$ and $L_j \oplus \sigma$, so label $L_i \oplus \sigma$ dominates label $L_j \oplus \sigma$. Hence, for any partial path $\mathscr{p}$ that $L_j \oplus \sigma$ can subsequently access and reach the destination, $L_i \oplus \sigma$ can also access it, and $\overline{c}(L_i \oplus \sigma \oplus \mathscr{p}) \le \overline{c}(L_j \oplus \sigma \oplus \mathscr{p})$ holds. This means that for any subsequent feasible extension of label L_j, label L_i can perform the same extension at a cost no greater than L_j's. Therefore, label L_i dominates label L_j, and **Proposition 1** is satisfied.

Appendix C.2. Proof of Proposition 2

Proof. Given two labels L_i^1 and L_i^2 with the current node i, they satisfy conditions (44), (45), (46), (48), (49), and (58). Let $L_i^1 \oplus j$ denote the label generated by extending L_i^1 to node j. For any extension node $j \in \mathcal{O}'(L_i^2)$, we can deduce:

$$w(L_i^1 \oplus j) = w(L_i^1) + w(j) \le w(L_i^2) + w(j) = w(L_i^2 \oplus j), \tag{C.7}$$

$$v(L_i^1 \oplus j) = v(L_i^1) + v(j) \le v(L_i^2) + v(j) = v(L_i^2 \oplus j), \tag{C.8}$$

$$t(L_i^1 \oplus j) = \max(\mathrm{e}_j, t(L_i^1) + \mathrm{s}_{ij}) \le \max(\mathrm{e}_j, t(L_i^2) + \mathrm{s}_{ij}) = t(L_i^2 \oplus j), \tag{C.9}$$

$$d(L_i^1 \oplus j) = d(L_i^1) + d_{ij} \le d(L_i^2) + d_{ij} = d(L_i^2 \oplus j), \tag{C.10}$$

$$\overline{c}(L_i^1 \oplus j) = \overline{c}(L_i^1) + \rho \cdot d_{ij} - \pi_{c(j)} \le \overline{c}(L_i^2) + \rho \cdot d_{ij} - \pi_{c(j)} = \overline{c}(L_i^2 \oplus j), \tag{C.11}$$

where ρ is taken as ρ_1 in the pricing problem for dedicated vehicles and as ρ_2 in the pricing problem for crowdsourced vehicles. Eqs. (C.7) − (C.11) ensure that $L_i^1 \oplus j$ is also feasible when $L_i^2 \oplus j$ is feasible.

From $\mathcal{O}'(L_i^1) \supseteq \mathcal{O}'(L_i^2)$, it follows that $\mathcal{O}'(L_i^1) \backslash \varsigma_j \backslash \{j\} \supseteq \mathcal{O}'(L_i^2) \backslash \varsigma_j \backslash \{j\}$. Since $\Psi(L_i^1 \oplus j)$ denotes the set of nodes that violate the time constraints upon extension from $L_i^1 \oplus j$, from (C.9), we can deduce $\Psi(L_i^1 \oplus j) \subseteq \Psi(L_i^2 \oplus j)$. Thus,

$$\mathcal{O}'(L_i^1 \oplus j) = \mathcal{O}'(L_i^1) \backslash \varsigma_j \backslash \{j\} \backslash \Psi(L_i^1 \oplus j) \supseteq \mathcal{O}'(L_i^2) \backslash \varsigma_j \backslash \{j\} \backslash \Psi(L_i^2 \oplus j) = \mathcal{O}'(L_i^2 \oplus j). \tag{C.12}$$

Therefore, $L_i^1 \oplus j$ and $L_i^2 \oplus j$ also satisfy the dominance rules (44), (45), (46), (48), (49), and (58).

Hence, by mathematical induction, it is easy to deduce that for any subsequent feasible extension of L_i^2, extending L_i^1 in the same manner remains feasible and incurs a cost less than or equal to that of L_i^2. **Proposition 2** is proved.

Appendix C.3. Proof of equivalence of Proposition 2 and Corollary 1

Proof. To establish the equivalence between **Proposition 2** and **Corollary 1**, it suffices to show that Eqs. (59) and (60) are equivalent to Eq. (61).

(1) **Proposition 2 implies Corollary 1.**

From Eq. (60), we have: $t(L_i) + s_{ij} + s_{jg} \leq \max\{e_j, t(L_i) + s_{ij}\} + s_{jg} \leq l_g$. This implies: $t(L_i) + s_{ij} \leq l_g - s_{jg}$. Additionally, from Eq. (59), we have: $t(L_i) + s_{ij} \leq l_j$. It yields: $t(L_i) + s_{ij} \leq \min\{l_g - s_{jg}, l_j\}$. Thus, Eq. (61) holds and **Proposition 2** implies **Corollary 1**.

(2) **Corollary 1 implies Proposition 2.**

When Eq. (61) holds, $t(L_i) + s_{ij} \leq l_j^* = \min\{l_g - s_{jg}, l_j\} \leq l_j$, so Eq. (59) holds. Also, $t(L_i) + s_{ij} \leq l_j^* = \min\{l_g - s_{jg}, l_j\} \leq l_g - s_{jg}$, accordingly, $t(L_i) + s_{ij} + s_{jg} \leq l_g$. Due to Rule 3, the extended node must satisfy $e_j + s_{jg} \leq l_g$, thus, $\max\{e_j, t(L_i) + s_{ij}\} + s_{jg} \leq l_g$, so Eq. (60) holds and **Corollary 1** implies **Proposition 2**.

Thus, **Proposition 2** and **Corollary 1** are equivalent.

Appendix C.4. Proof of Proposition 3

Proof. Suppose there exist labels L_{i^n} and L'_{i^k} satisfying the conditions of **Proposition 3**, where node $j \in \mathcal{O}'(L_{i^n})$ is the next node extended from L_{i^n}. By the conditions of **Proposition 3**, $\overline{c}(L_{i^k} \oplus j) \leq \overline{c}(L_{i^n} \oplus j)$. Since i^k is an intermediate node visited by L_{i^n}, and the distance calculation satisfies the triangle inequality, the travel distance for L_{i^n} to reach j along the visit sequence of $\mathcal{U}(L_{i^n})$ starting from node i^k must be no less than the distance of directly visiting j from i^k, this implies $d(L'_{i^k} \oplus j) \leq d(L_{i^n} \oplus j)$. Similarly, since the vehicle travel time is a multiple of the travel distance (and positive), $t(L'_{i^k} \oplus j) \leq t(L_{i^n} \oplus j)$. Moreover, because $\mathcal{U}(L'_{i^k}) \subset \mathcal{U}(L_{i^n})$, it follows straightforwardly that $\mathcal{O}'(L'_{i^k} \oplus j) \supseteq \mathcal{O}'(L_{i^n} \oplus j)$, $w(L'_{i^k} \oplus j) < w(L_{i^n} \oplus j)$, and $v(L'_{i^k} \oplus j) < v(L_{i^n} \oplus j)$. The above conclusions also ensure that $L'_{i^k} \oplus j$ satisfies the resource constraints. Therefore, $L'_{i^k} \oplus j$ dominates $L_{i^n} \oplus j$, and **Proposition 3** is proved.

Appendix C.5. Proof of Proposition 4

Proof. Suppose there exist labels L' and L that satisfy the conditions of **Proposition 4**. Since the two labels differ in only one visited node within the set of nodes visited by labels L' and L, i.e., nodes i_1^{n-1} and i_2^{n-1}. Nodes i_1^{n-1} and i_2^{n-1} belong to different locations of the same customer, thus $w(L') = w(L)$, $v(L') = v(L)$, $\pi_{c(i_1^{n-1})} = \pi_{c(i_2^{n-1})}$, so $\sum_{j \in \mathcal{U}(L')} \pi_{c(j)} = \sum_{j \in \mathcal{U}(L)} \pi_{c(j)}$. In addition, from Eqs. (62) and (63), it can be deduced that $t(L') \leq t(L)$, $d(L') < d(L)$, thus $\overline{c}(L') = \rho d(L') - \bar{\pi} - \sum_{j \in \mathcal{U}(L')} \pi_{c(j)} < \rho d(L) - \bar{\pi} - \sum_{j \in \mathcal{U}(L)} \pi_{c(j)} = \overline{c}(L)$, where ρ is equal to ρ_1 in the dedicated vehicle pricing problem and ρ_2 in the crowdsourced vehicle pricing problem; $\bar{\pi} = \pi_k$ if the current vehicle is a dedicated vehicle, and $\bar{\pi} = \rho_2 \cdot d_{a_m b_m} + \pi_m$ if it is a crowdsourced vehicle m.

Since $t(L') \leq t(L)$, it can be deduced that $\Psi(L') \subseteq \Psi(L)$, and the set of nodes that L cannot extend due to Rule 2 and Rule 3 contains the set of nodes that L' cannot extend due to these two rules. Moreover, because the only difference between $\mathcal{U}(L)$ and $\mathcal{U}(L')$ is that i_1^{n-1} is replaced by i_2^{n-1}, when only considering Rule 1, $\varsigma_{i_2^{n-1}} \cup \{i_2^{n-1}\} = \varsigma_{i_1^{n-1}} \cup \{i_1^{n-1}\}$. Thus, it can be easily deduced that

$\varsigma_{i_2^{n-1}} \cup \Psi(L') \cup \{i_2^{n-1}\} \subseteq \varsigma_{i_1^{n-1}} \cup \Psi(L) \cup \{i_1^{n-1}\}$. According to the update method for $\mathcal{O}'(L_i)$ (see Eq. (47) and **Proposition 2**), we know that $\mathcal{O}'(L') \supseteq \mathcal{O}'(L)$. **Proposition 2** is satisfied, and thus **Proposition 4** is proved.